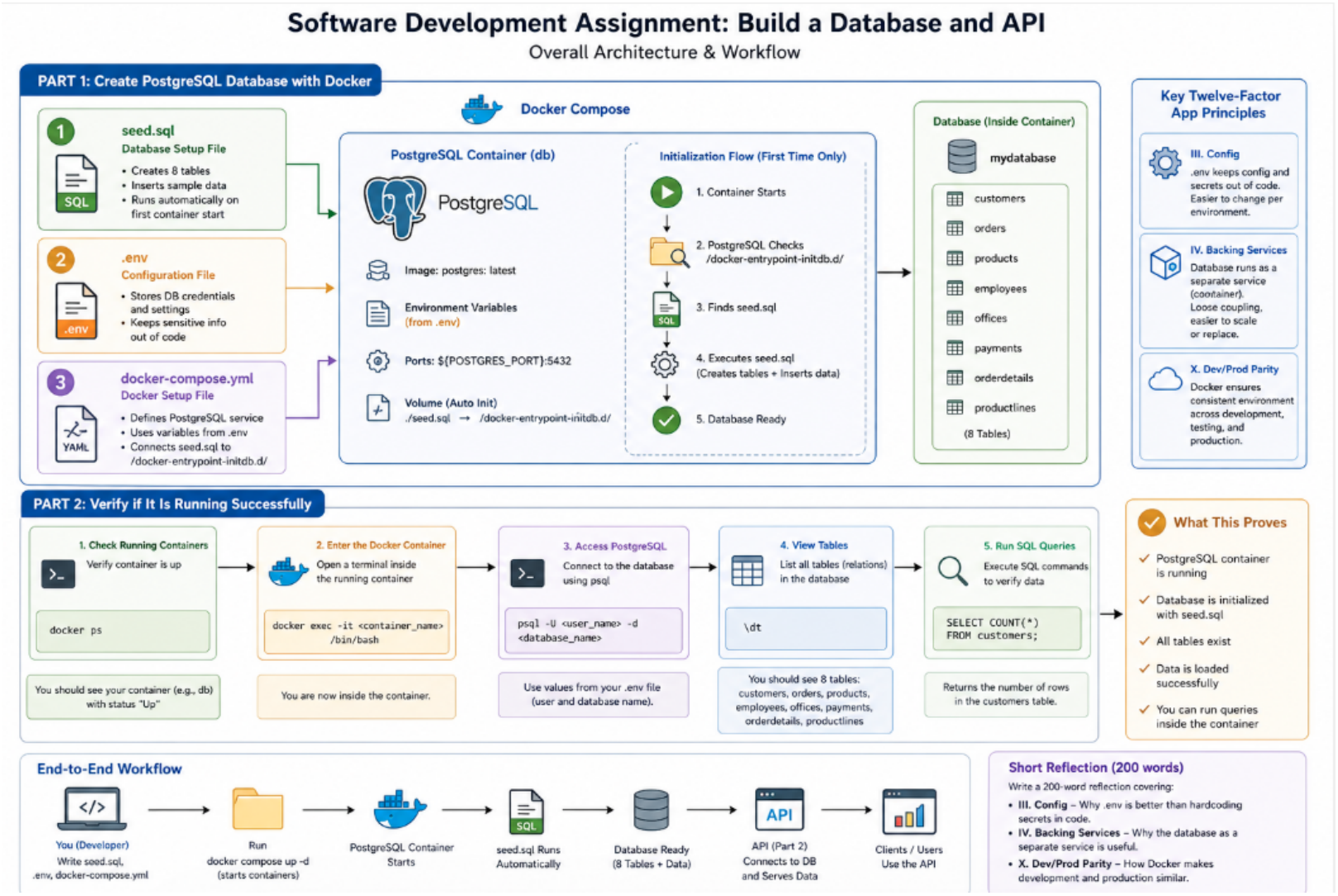
Software Development Assignment: Build a Database and API (Simple Guide)



This assignment has two main parts:

1. Create a database using Docker
2. Build an API to access the data

Part 1: Create a PostgreSQL Database with Docker

Goal

You will run a PostgreSQL database using Docker and automatically load data when it starts.

Step 1: Use seed.sql file which is provided.

1. seed.sql (Database Setup File)

- This file contains SQL statements used to initialize the database.
- It creates the required tables, including:
  - customers
  - orders
  - products
  - offices
  - employees
  - productlines
  - payments
  - orderdetails
- It inserts sample data into all the above tables.
- When the project is run using Docker Compose command, the seed.sql file should execute automatically.
- This ensures that the database is created, all tables are set up, and initial sample data is inserted without any manual steps.

Table	Rows
`productlines`	7
`products`	110
`offices`	7
`employees`	23
`customers`	122
`orders`	326
`payments`	273
`orderdetails`	2,996
Total	**3,864**

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2. .env (Set up configuration file)

- This file stores sensitive information like database username, password, hostname, port and so forth.
- Example:

POSTGRES_USER=admin
POSTGRES_PASSWORD=secret
POSTGRES_DB=mydatabase
POSTGRES_PORT=5432

- Do NOT write these values directly in your code.
- The provided example has placeholder values, do not literally use these values while setting up a postgres connection.

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3. docker-compose.yml (Setup File for Docker)

- This file tells Docker how to run PostgreSQL.
- Use variables from .env like this:

Follow these simple instructions:

- Create a service for your database (name it something like db).
- Use the official PostgreSQL image.

- Do **not** write username, password, or database name directly → use variables from .env like:
  ${POSTGRES_USER}

Important Setup Steps

- **Connect your SQL file**
  - You must link your seed.sql file to this folder inside the container:
    /docker-entrypoint-initdb.d/
  - This is the special location where PostgreSQL looks for setup scripts.

What This Does

- When you run your container for the first time:
  - PostgreSQL starts
  - It checks that folder
  - It finds seed.sql
  - It runs it automatically
  - Your tables and data are created

In simple words:
You are telling Docker:
"Start PostgreSQL, use my settings from .env, and run my SQL file automatically when the database is created."

Step 2: Short Reflection (200 words)

Write a simple explanation:

Config (Factor III)

- Why is .env better than writing passwords in code?

Backing Services (Factor IV)

- Why is treating the database as a separate service useful?

Dev/Prod Parity (Factor X)

- Why does Docker make development and production similar?

Part 2: Verify if it is running succesfully

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Once PostgreSQL starts successfully, your database should contain 8 tables, including the **customers** table.

After running Docker, you can verify that your container is up and running by using the appropriate command to check its status.

```
(base) fm-pc-lt-148@Acer-Predator:~/Desktop/AI_Fellowship/week2_tasks/fastapi_app$ docker ps
CONTAINER ID   IMAGE              COMMAND                  CREATED             STATUS
ORTS                                       NAMES
77ef99dec794   fastapi_app-api    "uvicorn main:app --…"   About an hour ago   Up About an hour
.0.0.0:8000->8000/tcp, [::]:8000->8000/tcp   fastapi_app-api-1
fa53fc83a46b   postgres:latest    "docker-entrypoint.s…"   About an hour ago   Up About an hour (healthy)
.0.0.0:5433->5432/tcp, [::]:5433->5432/tcp   fastapi_app-db-1
```

You can verify whether the container is running and confirm that PostgreSQL is operating in a separate container. Then, you can access that container to check if the tables have been properly populated using the available commands.

```
fa53fc83a46b   postgres:latest   "docker-entrypoint.s…"   About an hour ago   Up About an hour (healthy)   0.0.0.0:5433->5432/tcp,
[::]:5433->5432/tcp   fastapi_app-db-1
(base) fm-pc-lt-148@Acer-Predator:~/Desktop/AI_Fellowship/week2_tasks/fastapi_app$ docker exec -it fa53fc83a46b /bin/bash
root@fa53fc83a46b:/# psql -U postgres -d classicmodels
psql (18.3 (Debian 18.3-1.pgdg13+1))
Type "help" for help.

classicmodels=# \dt
              List of tables
 Schema |     Name     | Type  |  Owner
--------+--------------+-------+----------
 public | customers    | table | postgres
 public | employees    | table | postgres
 public | offices      | table | postgres
 public | orderdetails | table | postgres
 public | orders       | table | postgres
 public | payments     | table | postgres
 public | productlines | table | postgres
 public | products     | table | postgres
(8 rows)

classicmodels=# select count(*) from customers;
 count
-------
   122
(1 row)
```

1. Enter the Docker Container

Use this command to open a terminal inside your running container:

<code>docker exec -it <container_name> /bin/bash</code>

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2. Access PostgreSQL

Once inside the container, connect to the database using:

<code>
psql -U <user_name> -d <database_name>
</code>

- Use the same **user_name** and **database_name** that you defined in your .env file.

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3. View Tables in the Database

To see all tables (relations), run:

<code>\dt</code>

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4. Run SQL Queries

You can now execute SQL commands.

Example: Count number of rows in the customers table:

<code>SELECT COUNT(*) FROM customers;</code>

Simple Summary

1. Enter container
2. Connect to PostgreSQL
3. List tables
4. Run queries